正面第 1 栏 | 基础 / 数学

题干定位： 定义查 F01/F02；维度查 M01；求导查 M02；方程组查 M03。

F01 Intelligent systems; AI V.S. CI

An intelligent system exhibits learning, adaptation, inference or planning. **Deduction** applies rules to facts; **induction** generalizes from examples and needs unseen-data checks. AI includes learning and rule-based methods. CI emphasizes neural learning, fuzzy representation and evolutionary optimization; they overlap. Traditional hard computing emphasizes precision; soft computing tolerates uncertainty. Classification predicts a class; regression predicts a number. Associative memory retrieves patterns from partial/noisy cues.

F02 ANN: architecture, rules and learning

ANN = (ARCH, RULE), ARCH=(u, v, w, x, y): $u, v \in \mathbb{R}^N$ are net inputs/activations; $w \in \mathbb{R}^M$ stores connection parameters; $x \in \mathbb{R}^d$, $y \in \mathbb{R}^m$ are external inputs/outputs.
G aggregates: $u = Wx + b$. **F** activates: $v = f(u)$. **H** reads out: $y = h(v)$. **E** evaluates, e.g. $\frac{1}{2}\|z - y\|^2$. **L** updates parameters. 按 G→F→H 计算。Training adds E→L.
Supervised learning uses targets z; unsupervised learning finds structure without labels. Training changes parameters; inference fixes them. States u, v change fast; weights change slowly.
Classification axes: deterministic/stochastic activation; feedforward/recurrent connections; supervised/unsupervised learning. Semilinear aggregation uses weighted sums; higher-order aggregation adds input products.
Discrete: $w_{t+1} = w_t + L$; continuous: $\dot{w} = L$, $\Delta w \approx \Delta t L$. NNs can offer nonlinear mappings, adaptive weights and distributed knowledge.

M01 Dimensions; linear V.S. affine

使用列向量。$w^T x = \sum_j w_j x_j \in \mathbb{R}$; $A \in \mathbb{R}^{m\times d}$ gives $Ax \in \mathbb{R}^m$. 先核对内维；$(AB)^T = B^T A^T$, usually $AB \neq BA$.
Ax is linear; $Ax + b$ is affine, linear only if $b = 0$. 矩阵乘法按行乘列计算：$(AB)_{ij} = \sum_k A_{ik} B_{kj}$.
微例： 给定 $w = (2, -1)^T$, $x = (1, 3)^T$, $b = 2$：代入算 $u = 2(1) - 1(3) + 2 = 1$，再得 $H(u) = 1$。

M02 Derivatives; gradients; extrema

基本导数：$(u^p)' = pu^{p-1}$, $(e^u)' = e^u$, $(\ln u)' = 1/u$ $(u > 0)$. 链式求导：沿同一路径把导数相乘，多条分支的贡献相加。For $a = a(\theta)$, $\partial_\theta \frac{1}{2}(z - a)^2 = (a - z)\partial_\theta a$. 按 $\theta^+ = \theta - \eta \nabla E$ 做梯度下降，$\eta > 0$。

$$\nabla_w \tfrac{1}{2}\|Xw - y\|^2 = X^T(Xw - y),$$
$$\nabla_w (w^T A w) = (A + A^T)w.$$

For symmetric A, $\nabla \frac{1}{2} w^T A w = Aw$; $\nabla \frac{1}{2}\|w\|^2 = w$. 先解 $\nabla E = 0$。Scalar $E'' > 0$: local minimum; $E'' < 0$: local maximum; zero: inconclusive. Multivariable Hessian $\nabla^2 E$: positive definite gives a local minimum; negative definite a local maximum; indefinite a saddle; semidefinite is inconclusive. Convex least squares has global minima; full column rank gives uniqueness.
微例： 设 $E = \frac{1}{2}(1 - 2w)^2$：求导得 $E' = 4w - 2 = 0 \Rightarrow w = 1/2$；再查 $E'' = 4 > 0$。

M03 消元 / 回代 / 检查

先将未知量按顺序排入 q，写成 $Aq = c$。对 $[A|c]$ 的**整行**交换、乘非零常数或加另一行的倍数；消元后回代，再查原方程。
微例： 给定 $u + v = 3$, $2u - v = 0$：做 $R_2 - 2R_1$ 得 $-3v = -6$；回代得 $v = 2$, $u = 3 - 2 = 1$。
$[0 \ \cdots \ 0|c \neq 0]$ means no solution. 记 $r = \text{rank}\, A$, $r_a = \text{rank}[A|c]$, $n = \dim q$: $r \neq r_a$ gives no solution; $r = r_a = n$ gives one; $r = r_a < n$ gives infinitely many (free unknowns).

正面第 2 栏 | 逻辑 / Perceptron

N01 M-P veto; gates; decoders

M-P inputs/outputs are 0/1. **先查抑制**: Any active inhibitory input forces output 0. **否则数兴奋输入** s: The output is 1 iff $s \geq \theta$。Excitation cannot cancel veto.
An ordinary TLU uses $u = w^T x - \theta = w^T x + b$, $b = -\theta$, and $H(u) = 1$ iff $u \geq 0$; negative weights only reduce the sum.
微例： 给定 $x = (1, 1, 1)$, $w = (1, 1, -1)$, $\theta = 1$：普通 TLU 的加权和 $= 1 \Rightarrow 1$；若 M-P 的第 3 路为抑制，得 0。AND uses n excitatory inputs, $\theta = n$; OR uses $\theta = 1$. NOT uses one inhibitory input, $\theta = 0$; NOR uses all inhibitory inputs, $\theta = 0$.
Decoder 构造： 对目标模式 r，1 位接兴奋、0 位接抑制，取 $\theta = \sum_j r_j$。普通 TLU 取 $w_j = 2r_j - 1$，活用同一 θ。Every mismatch lowers the score by 1.
微例： 目标 101：取 $w = (1, -1, 1)$, $\theta = 2$。输入 101 得分 2，输出 1；输入 111 得分 1，输出 0。
构造： 为各输出 1 的行建 decoder，再接 OR。Any Boolean function can be implemented with two computational layers. Uninhibited networks are monotone: increasing inputs cannot turn 1 into 0. One unit need not represent all monotone functions. **注意：** Positive scaling of both w, θ preserves predictions.

N02 Feedforward V.S. recurrent

A feedforward network has no feedback cycle; a recurrent network uses old state and current input. Both compute through connected units. Recurrent M-P networks can simulate any finite automaton (finite state/input sets). **同步计算：** 先写初态和时序；固定旧状态；算完全部新状态后统一替换。

D01 M-P V.S. Perceptron

Both threshold aggregated inputs. M-P uses fixed connections/veto for logic design; a perceptron learns real weights/bias from labels for linear classification, without veto. Neither single unit represents XOR.

P01 Perceptron: 逐样本更新

Inputs and targets satisfy $x \in \mathbb{R}^d$, $z \in \{0, 1\}$。先算旧 $u = w^T x + b$、$\hat{y} = H(u)$ $(H(0) = 1)$，再算 $e = z - \hat{y}$:

$$w^+ = w + \eta e x, \qquad b^+ = b + \eta e.$$

按样本顺序计算，下一样本用新参数；每轮后重查全部样本。增广为 $\tilde{x} = [x; 1]$、$\tilde{w} = [w; b]$ 可一起更新。
微例： 初值 $w = (0, 0)$, $b = 0$, $\eta = 1$。OR 样本 00, $z = 0$：算得 $u = 0$, $e = -1$，更新得 $b = -1$。下一样本 01, $z = 1$：算得 $u = -1$, $e = 1$，更新得 $w = (0, 1)$, $b = 0$。
注意： 标签为 ± 1 时，错分的 $z - \hat{y} = \pm 2$；若题目规定只在错分时加 $\eta z x$，注意步长约定不同。若 $b = -\theta$，按 $\theta^+ = \theta - \eta e$ 更新阈值。
For bipolar inputs $s = 2x - 1$, equivalent parameters are $w_s = w/2$, $b_s = b + \frac{1}{2}\sum_j w_j$.

P02 Separability; weight space; counts

Standard perceptron learning makes finitely many corrections on a finite linearly separable set; it does not maximize margin. One hyperplane cannot separate XOR/XNOR. Fixed parameters define an input-space boundary; a fixed sample defines the weight-space boundary $\tilde{w}^T \tilde{x}_i = 0$. This is geometric, not SVM Lagrangian, duality.
d binary inputs give 2^d rows and 2^{2^d} functions. For m central hyperplanes in general position in $\mathbb{R}^D$, $D = d + 1$:

$$R = 2\sum_{k=0}^{D-1} \binom{m-1}{k}.$$

Out-of-range binomials are zero; degeneracy can reduce R. 代入 $m = 4$, $D = 3$，得 $R = 2(1 + 3 + 3) = 14$。14/16 two-input functions are threshold functions.

正面第 3 栏 | LMS / 前向计算

题干定位： 类别误差查 P01；线性残差查 A01；多层梯度查 B02。先核对输出和损失。

A01 ADALINE; LMS; least squares

ADALINE (adaptive linear element) trains **linear output** $a = w^T x + b$ with $e = z - a$. 先算训练残差，仅在分类输出时阈值化。本处取 $E = \frac{1}{2}\sum_i e_i^2$:

$$\nabla_w E = -\sum_i e_i x_i, \qquad \partial_b E = -\sum_i e_i.$$

Online 逐样本: 算一个残差，更新 $w^+ = w + \eta e x$、$b^+ = b + \eta e$. **Batch 整批:** 固定旧参数，算全部残差并累加更新量，最后统一更新一次。均值损失的梯度再除样本数。
微例： 给定 $x = (1, 2)$, $z = 1$, $w = (.1, -.1)$, $b = 0$、$\eta = .1$：算 $a = .1 - .2 = -.1$; $e = 1.1$; $\Delta w = .1(1.1)(1, 2) = (.11, .22)$；更新得 $w^+ = (.21, .12)$, $b^+ = .11$。回代查 $a^+ = .21 + .24 + .11 = .56$。
代数求解： 将第 i 个样本写成 X 的行 $[x_i^T, 1]$，取 $\beta = [w; b]$，解 $X^T X \beta = X^T z$。Full column rank gives uniqueness; rank deficiency requires free variables or a pseudoinverse.
微例： XOR 各行为 00, 01, 10, 11，目标 $(-1, 1, 1, -1)$，代入得 $X^T z = 0$。利用满列秩得 $\beta = 0$。The best linear square fit still cannot classify XOR.
注意： 目标没有 1/2 时，梯度乘 2。若 $a = \kappa(w^T x + b)$，求导时别漏因子 κ。

D02 Perceptron V.S. ADALINE

共性： Both learn one affine layer from labels for linear classification. **差异：** A perceptron corrects **hard class errors**; ADALINE minimizes **linear-output squared error**, supporting regression or thresholded classification.
限制： Perceptron convergence requires separability and does not guarantee maximum margin. ADALINE's linear least-squares objective is convex, but minimizing it need not minimize classification errors. Neither single unit represents XOR. Online/batch specify update timing, not exclusive models.

B01 MLP; activation / 前向计算

An MLP is a model with nonlinear hidden units and an output layer. BP computes gradients; it is a training method, not a competing model. Hidden units normally have no direct labels.
先取 $a^0 = x$。For layer l, $W^l \in \mathbb{R}^{m_l \times m_{l-1}}$, $b^l, u^l, a^l \in \mathbb{R}^{m_l}$:

$$u^l = W^l a^{l-1} + b^l, \qquad a^l = f_l(u^l).$$

前向计算： 先列维度；算净输入；代入激活函数；逐层继续；最后算输出和损失。
Sigmoid $a = (1 + e^{-u})^{-1}$ has $a' = a(1 - a)$.
Tanh $a = (1 - e^{-2u})/(1 + e^{-2u})$ has $a' = 1 - a^2$. For $a = \sigma(cu)$, $a' = c\,a(1 - a)$. For $a = (1 - e^{-u})/(1 + e^{-u}) = \tanh(u/2)$, $a' = \frac{1}{2}(1 - a^2)$. Saturation gives small derivatives. 分段激活的折点需写明导数约定。
Affine layers alone remain affine. Nonlinear hidden units can represent XOR: $h_1 = H(x_1 - x_2 - .5)$, $h_2 = H(-x_1 + x_2 - .5)$, $\hat{y} = H(h_1 + h_2 - .5)$.
微例： 代入 $x = (1, 0)$，净输入为 $(.5, -1.5)$，隐藏输出 $(1, 0)$，输出层净输入 $.5$，故输出 1。代入 $(1, 1)$，隐藏输出 $(0, 0)$，故输出 0。
注明层数是否计输入层；按实际跳连计算。

正面第 4 栏 | BP / 泛化

B02 BP: 逐单元求梯度与更新

答句： Backpropagation (BP, generalized delta rule) computes gradients by the chain rule. An optimizer uses them to update parameters; BP is not a network architecture.
本处取 $E = \frac{1}{2}\sum_k (z_k - o_k)^2$，$z_k$ 代入目标，o_k 代入输出，取 $\delta_j = -\partial E/\partial u_j$。

$$\delta_k = (z_k - o_k) f_k'(u_k) \quad \text{(output)},$$
$$\delta_j = f_j'(u_j) \sum_k v_{kj}\delta_k \quad \text{(hidden)},$$
$$\frac{\partial E}{\partial w_{ji}} = -\delta_j a_i, \qquad \frac{\partial E}{\partial b_j} = -\delta_j,$$
$$w_{ji}^+ = w_{ji} + \eta\delta_j a_i, \quad b_j^+ = b_j + \eta\delta_j.$$

a_i 代入前层输出（首个可训练层用 x_i）；v_{kj} 取**从 j 到 k 的旧权重**。For sigmoid $f' = a(1 - a)$; for a linear output $f' = 1$.
计算步骤： (1) 固定全部旧参数；(2) 前向算完并保存各 u, a；(3) 算输出层 delta；(4) 逐隐藏层向后算，**全部下游贡献相加**；(5) 列出所有梯度；(6) 统一更新全部参数一次；(7) 重新计算输出和损失。**注意：** 隐藏 delta 用下游**旧权重**；原始输入不另求 sigmoid 导数。若题目将 delta 定义为正导数，更新时改用减号。
微例： Sigmoid 隐藏 / 输出单元；给定 $x = z = \eta = 1$，输入至隐藏层 $(w, b_h) = (0, 0)$，隐藏至输出层 $(v, b_o) = (2, -1)$。
$u_h = wx + b_h = 0$, $h = .5$; $u_o = vh + b_o = 0$, $o = .5$; $E = 1/8$.

$$\delta_o = (1 - .5)(.5)(.5) = 1/8,$$
$$\delta_h = (.5)(.5)2(1/8) = 1/16.$$

梯度按 (w, b_h, v, b_o) 顺序排列为 $(-1/16, -1/16, -1/16, -1/8)$。代入更新式得

$$(w^+, b_h^+, v^+, b_o^+) = (1/16, 1/16, 33/16, -7/8).$$

回代检查 $h^+ = \sigma(1/8)$ 及 $o^+ = \sigma((33/16)h^+ - 7/8) \approx .5549$，得 $E^+ \approx .0990$。
分支检查： 给定 $h = .5$、下游权重 $(2, -1)$ 及 delta $(.1, .2)$，代入得 $\delta_h = .25[2(.1) - 1(.2)] = 0$。
Batch 计算： 各样本用同一旧参数算梯度，累加后统一更新；均值损失再除 n。Inference uses only the forward pass.

B03 Momentum / 模型判断

$$\Delta\theta_t = -\eta g_t + \mu\Delta\theta_{t-1}, \quad 0 \leq \mu < 1,$$
$$\theta_{t+1} = \theta_t + \Delta\theta_t.$$

计算步骤： 将旧**增量**乘 μ，再加 $-\eta g_t$。最后把新增量加到 θ_t；不要把旧参数当增量。
微例： 给定 $g = .3$, $\eta = .1$, $\mu = .8$, $\Delta\theta_{t-1} = .02$：代入得 $\Delta\theta = -.03 + .016 = -.014$。
答句： Momentum can reduce oscillation, but guarantees neither a global minimum nor loss reduction at every step.
Representation concerns whether suitable parameters exist; optimization concerns finding them; generalization concerns unseen-data performance. One does not guarantee another. Kolmogorov's representation is an existence result using univariate functions and sums, not a training guarantee. With suitable activation and sufficient capacity, networks approximate continuous functions on compact domains arbitrarily well. This does not guarantee a fixed small network or successful BP training.
Overfitting means fitting training details while predicting unseen data poorly. Falling training loss with rising validation loss suggests overfitting. Validation data guide capacity and stopping-time selection; test data are reserved for final evaluation. A stationary point of a nonconvex loss need not be a global minimum.

正面第 5 栏 | RBF、FLN、ELM

R01 RBF: 计算特征与输出权重

答句： A radial basis function depends only on distance from its center. A Gaussian RBF network uses radial hidden features and a linear output:

$$h_j(x) = e^{-\|x - c_j\|^2/s_j^2}, \quad s_j > 0,$$
$$f(x) = \sum_j \beta_j h_j(x) + b.$$

Centers set response locations; larger widths give slower decay. Other radial bases include r^k $(k = 1, 3, \ldots)$, $r^2 \log r$ (limit 0 at $r = 0$), $\sqrt{r^2 + s^2}$ and its reciprocal; not all are local.
计算步骤： 固定中心 / 宽度；算**平方**距离；核对分母；令 $H_{ij} = h_j(x_i)$；只在拟合输出 bias 时加一列 1；解系数后回代检查。Each row is one sample. With n samples and h features, $H \in \mathbb{R}^{n\times h}$ and $\beta \in \mathbb{R}^h$ before adding bias.
最小二乘时解 $H^T H\beta = H^T y$；特征秩亏时用伪逆，不要默认 H^{-1} 存在。
微例： 无 bias；从第 2 行消去 β_1：

$$\left[\begin{array}{cc|c} 1 & 1/2 & 1 \\ 1/2 & 1 & 0 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - \frac{1}{2}R_1} \frac{3}{4}\beta_2 = -\frac{1}{2}.$$

回代得 $\beta_2 = -2/3$、$\beta_1 = 1 - \beta_2/2 = 4/3$。
XOR 代入： 中心 $c_1 = (1, 1)$、$c_2 = (0, 0)$，宽度 1：

$$00 \mapsto (e^{-2}, 1), \quad 11 \mapsto (1, e^{-2}),$$
$$01, 10 \mapsto (e^{-1}, e^{-1}).$$

取 $g = .9 - h_1 - h_2$，XOR 正类点得 $.164 > 0$，负类点得 $-.235 < 0$。
Cover / ϕ-separability: The condition is $w^T\phi > 0$ for positive points and $w^T\phi < 0$ for negative points; bias can be augmented. Suitable nonlinear features may allow separation; arbitrary mappings guarantee neither separation nor generalization. **注意：** 按题设使用 s^2 或 $2s^2$ 分母。Optimizing centers/widths changes the fixed-feature linear problem.

R02 FLN / ELM: 拟合哪些参数

答句： FLN prescribes functional expansions and fits output coefficients. Basic ELM randomly initializes and fixes hidden weights/biases, then fits output weights. ELM has one hidden layer; it does not lack hidden units.
计算步骤： 固定 ELM 隐藏参数，算 $H_{ij} = f(a_j^T x_i + b_j)$，再解 $H\beta \approx y$；预测时沿用同一组特征。
微例： FLN 取 $\psi = (1, x_1, x_2, x_1x_2)$、$\beta = (0, 1, 1, -2)$，代入得 XOR $f = x_1 + x_2 - 2x_1x_2$。
微例： ELM 给定 $h_1 = \sigma((\log 3)x)$、$h_2 = \sigma(-(\log 3)x)$，数据 $(x, y) = (0, 0), (1, 1)$，输出无 bias。H 的两行为 $(1/2, 1/2)$ 和 $(3/4, 1/4)$。联立 $\beta_1 + \beta_2 = 0$、$3\beta_1 + \beta_2 = 4$，得 $\beta = (2, -2)^T$。回代查 $H\beta = (0, 1)^T$。

D03 MLP V.S. RBF

Both are feedforward models for classification or regression. MLP applies activations to affine weighted sums and usually trains layers jointly with BP. Gaussian RBF uses distances to centers and commonly fixes centers/widths before fitting linear outputs. Joint training is generally nonconvex; fixed-feature squared-output fitting is convex. Neither is always faster or more accurate.

D04 RBF V.S. FLN V.S. ELM

All can fit linear outputs on fixed nonlinear features. RBF uses distances/centers/widths; FLN uses prescribed functions; ELM uses random fixed hidden parameters. All support classification/regression. Fixed-feature least squares has no bad local minima, but uniqueness and good generalization are not guaranteed.

背面第 1 栏 | 间隔 / KKT

S01 SVM: maximum margin and primal

题干定位： margin / hyperplane / maximum / separable → S01; dual / support vector / KKT → S02.

答卷句： Hard SVM selects a hyperplane with the maximum geometric margin. Hard margin requires separability in the chosen feature space. It can be outlier-sensitive and does not guarantee the best test performance.

模型： $\phi_i,\ w\in\mathbb{R}^h,\ b\in\mathbb{R}$, labels $y_i=\pm1$.

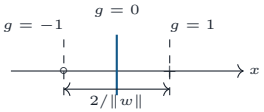
$$g(x)=w^T\phi(x)+b,\qquad \hat{y}=\operatorname{sign} g(x).$$

A linear SVM uses $\phi(x)=x$. The decision boundary is $g=0$; 此时按题目给定约定判类。$g>0$ predicts $+1$; $g<0$ predicts -1. The weight vector is normal to the hyperplane; the bias shifts its position. With bias, the score is an affine function. **Functional margin:** $m_i=y_ig_i$. **Geometric margin:** $\rho_i=m_i/\|w\|$. For $w\neq0$, distance is $|g|/\|w\|$. Positive scaling of (w,b) preserves geometry/classification, but changes functional margins.

规范化：将最近训练样本的函数间隔设为 $m=1$。The support planes are $g=\pm1$; each lies $1/\|w\|$ from the boundary.

$$\max\ \frac{2}{\|w\|}\iff\boxed{\min_{w,b}\ \frac{1}{2}\|w\|^2},\qquad m_i\ge1.$$

Total width: $2/\|w\|$ under this normalization. The primal minimizes squared norm; it maximizes geometric margin. Unlike a perceptron, SVM optimizes a margin criterion rather than accepting any separating boundary.



操作： 列出每个 $y_ig_i\ge1$；联立紧约束求 w,b；回代全体样本；再算宽度；用 KKT 确认最优。**微例：** $(-1,-1),(1,+1),\ g=wx+b$: $w-b\ge1$, $w+b\ge1$, 得到 $w\ge1+|b|$。最优解：$w=1,b=0$; 总宽 2；原始目标值 $1/2$。

S02 Dual, KKT and prediction

答卷句： Support vectors have nonzero dual coefficients and determine prediction. KKT combines feasibility, stationarity and complementary slackness. A margin point may have a zero coefficient.

$$L=\tfrac12\|w\|^2+\sum_i\alpha_i(1-y_ig_i),\qquad \alpha_i\ge0.$$

The Lagrangian is minimized over w,b and maximized over α.

$$\partial_wL=0:\ \ w=\sum_i\alpha_iy_i\phi_i,$$

$$\partial_bL=0:\ \ \sum_i\alpha_iy_i=0.$$

$$\max_\alpha D=\sum_i\alpha_i-\frac12\sum_{i,j}\alpha_i\alpha_jy_iy_jK_{ij},$$

约束：$\alpha_i\ge0,\ y^T\alpha=0$; $K_{ij}=\phi_i^T\phi_j$。**逐项检查：** $m_i\ge1;\ \alpha_i\ge0$; 上面两条驻点式；$\alpha_i(m_i-1)=0$.

$$m_i>1\Rightarrow\alpha_i=0,\qquad \alpha_i>0\Rightarrow m_i=1.$$

操作： 先求 α；选 $\alpha_s>0$ 求 b；多点核对；再计算新点分数。

$$b=y_s-\sum_i\alpha_iy_iK(x_i,x_s),$$

$$\boxed{g(x)=\sum_i\alpha_i\boxed{y_i}K(x_i,x)+b}.$$

微例： 接 S01: $\alpha_1=\alpha_2=a,\ D=2a-2a^2$. $D'=2-4a=0\Rightarrow a=1/2$; $D''=-4<0$. 对偶值 = 原始值 $=1/2$。

注意： 分类预测中保留 y_i。Soft-margin bias recovery from the boundary equality requires $0<\alpha_s<C$.

背面第 2 栏 | 软间隔 SVM / 核

S03 Soft margin: slack and C

答卷句： Soft-margin SVM permits penalized margin violations. Slack does not mean misclassification. Larger C penalizes violations more heavily, without guaranteeing lower test error.

$$\min\ \tfrac12\|w\|^2+C\sum_i\xi_i,\qquad C>0,$$

$$m_i=y_ig_i\ge1-\xi_i,\qquad \xi_i\ge0.$$

操作： 先算 $m_i=y_ig_i$，再逐点取最小可行松弛：最小松弛：

$$\boxed{\xi_i=\max(0,1-m_i)}\ \text{(hinge loss)}.$$

For minimum slacks: $m\ge1$: no violation; $0<m<1$: correct but inside the margin; $m=0$: decision boundary; $m<0$: misclassified, $\xi>1$. **微例：** $m=(1.4,.4,0,-.3)\Rightarrow\xi=(0,.6,1,1.3)$. 代入目标：$J=\|w\|^2/2+C\sum\xi$。

沿用 S02 的对偶目标，并加 $0\le\alpha_i\le C$ 及 $y^T\alpha=0$。

$$\alpha_i(m_i-1+\xi_i)=0,\qquad (C-\alpha_i)\xi_i=0.$$

最优乘子 α	必须满足
0	$m\ge1$
$0<\alpha<C$	$m=1,\ \xi=0$
C	$m\le1$

$\alpha=C$ does not imply misclassification; $m=1$ does not determine α.

K01 Kernel / Gram: 答句与计算

答卷句： A kernel evaluates feature-space inner products without constructing features: $K(x,z)=\phi(x)^T\phi(z)$. The classifier is affine in feature space.

核函数： linear x^Tz; polynomial $(1+x^Tz)^p$ for a nonnegative integer p; Gaussian

$$K(x,z)=\exp\left[-\frac{\|x-z\|^2}{2\sigma^2}\right],\qquad \sigma>0.$$

Sigmoid $\tanh(\kappa x^Tz+\theta)$ requires valid parameters/domain. Denominators σ^2 and $2\sigma^2$ use different width conventions.

有效性： All finite Gram matrices must be symmetric PSD: $c^TKc\ge0$ for all c. One dataset cannot prove global validity. **微例：** $G=\left[\begin{smallmatrix}1&2\\2&1\end{smallmatrix}\right]$, $c=(1,-1)^T$: $c^TGc=-2<0$; 该 Gram 矩阵无效。

For two-dimensional inputs, $(1+x^Tz)^2$ admits

$$\phi(x)=(1,\sqrt2x_1,\sqrt2x_2,x_1^2,\sqrt2x_1x_2,x_2^2)^T.$$

操作： 按样本顺序算 K_{ij}；检查对称和对角；代入 dual；新点算 $K(x_i,x)$；乘 α_iy_i 求和、加 b、取 sign。**微例：** Gaussian 核，$x=(0,1),\sigma=1$: 对角为 1，非对角为 $e^{-1/2}$。给定 $\alpha=(1,1),y=(-1,1),b=0$，代入得 $g(0)=-1+e^{-1/2}<0$。

Model selection uses validation data to choose C/kernel/width. Small σ may overfit.

D05 模型比较答句

Hard-margin V.S. Soft-margin SVM. Both regularize norm and classify by margin. Hard margin requires $m_i\ge1$ and separable data; it can be infeasible/outlier-sensitive. Soft margin uses penalized slacks and $\alpha_i\le C$ for noisy/overlapping data. Both need parameter selection.

RBF network V.S. RBF-kernel SVM. Both use radial similarities. RBF networks select/learn centers and widths, then often fit linear outputs for classification/regression. RBF-kernel SVM uses training samples as centers and a margin-based QP for constrained α_iy_i. Its expansion is often sparse, possibly with many SVs. Similar bases do not imply identical training. Both need feature/kernel choices. SVR handles regression.

背面第 3 栏 | SVR / 偏置

V01 Support vector regression

答卷句： SVR predicts continuous targets. The ε-insensitive loss ignores errors inside the tube and linearly penalizes excess errors. Norm controls flatness; C weights the penalty. Full output height: 2ε. ε sets the tolerated output error; C sets its excess-error cost. They are different parameters. Prediction returns a number, not a class sign.

$$f_i=w^T\phi_i+b,\quad r_i=y_i-f_i,\quad y_i\in\mathbb{R},$$

$$\ell_\varepsilon(r)=\max(0,|r|-\varepsilon),\quad \varepsilon\ge0.$$

Hard tube: $\min\|w\|^2/2$ subject to $|r_i|\le\varepsilon$. Soft tube minimizes J over w,b,ξ,ξ^*:

$$J=\tfrac12\|w\|^2+C\sum_i(\xi_i+\xi_i^*),\qquad C>0,$$

$$r_i\le\varepsilon+\xi_i,\quad -r_i\le\varepsilon+\xi_i^*,\quad \xi_i,\xi_i^*\ge0.$$

操作： 统一 $r=y-f$；算两侧最小松弛；最后相加计罚：$\xi_i=\max(0,r_i-\varepsilon)$; $\xi_i^*=\max(0,-r_i-\varepsilon)$. ξ corresponds to underprediction; ξ^* to overprediction. **微例：** $y=3,f=2,\varepsilon=.4$: $r=1,\xi=.6,\xi^*=0$, 损失 .6。

构造 L： 取 $\alpha_i,\alpha_i^*,\eta_i,\eta_i^*\ge0$.

$$\begin{aligned}L=J&+\sum_i\alpha_i(r_i-\varepsilon-\xi_i)\\&+\sum_i\alpha_i^*(-r_i-\varepsilon-\xi_i^*)\\&-\sum_i(\eta_i\xi_i+\eta_i^*\xi_i^*).\end{aligned}$$

操作： 对 w,b,ξ,ξ^* 求导置零，消元：

$$d_i=\alpha_i-\alpha_i^*,\quad w=\sum_id_i\phi_i,\quad \sum_id_i=0,$$

松弛变量求导后整理：$\eta_i=C-\alpha_i$, $\eta_i^*=C-\alpha_i^*$。

$$\max_{\alpha,\alpha^*}\ -\tfrac12d^TKd-\varepsilon\mathbf{1}^T(\alpha+\alpha^*)+y^Td,$$

$$\mathbf{1}^Td=0,\qquad 0\le\alpha_i,\alpha_i^*\le C.$$

预测：$f(x)=\sum_id_iK(x_i,x)+b$。ε 项用两乘子的**和**，预测系数用两乘子的**差**；回归预测不要再乘 y_i。

V02 求偏置与 KKT 检查

$$\alpha_i(\varepsilon+\xi_i-r_i)=0,\quad \alpha_i^*(\varepsilon+\xi_i^*+r_i)=0,$$

$$(C-\alpha_i)\xi_i=(C-\alpha_i^*)\xi_i^*=0.$$

先算 $q_i=\sum_jd_jK(x_j,x_i)$。**Free SV:** $0<\alpha_i^{(*)}<C$. If $|r_i|<\varepsilon$, both multipliers are zero.

$$0<\alpha_i<C:\quad b=y_i-q_i-\varepsilon,$$

$$0<\alpha_i^*<C:\quad b=y_i-q_i+\varepsilon.$$

操作： 选自由乘子对应的样本求 b；其他自由样本核对同值；再回代残差、松弛与全部 KKT。**微例：** 线性核，$x=y=(0,2),\varepsilon=1/2,C=1,\ d=(-1/4,1/4),\ \alpha_2=1/4$: 算得 $w=1/2,\ q_2=1,\ b=2-1-1/2=1/2$。残差 $(-1/2,1/2)$；两侧松弛均为零。

无自由 SV (no free SV): 对全部适用界取交集：

$$\begin{aligned}\alpha_i=0:&\quad b\ge y_i-q_i-\varepsilon,\\ \alpha_i=C:&\quad b\le y_i-q_i-\varepsilon,\\ \alpha_i^*=0:&\quad b\le y_i-q_i+\varepsilon,\\ \alpha_i^*=C:&\quad b\ge y_i-q_i+\varepsilon.\end{aligned}$$

操作： 取最大下界 $\le b\le$ 最小上界；交集为空则重查系数；选区间内 b 回代。**微例：** $x=(0,1),y=(0,2),\varepsilon=C=1/2$, $d=(-1/2,1/2)$, 两非零乘子都等于 C: $q=(0,1/2)$, $b\in[1/2,1]$。取 $b=3/4$，得到 $r=(-3/4,3/4)$；每点一侧松弛为 $1/4$。

注意： For $\varepsilon>0$, one sample cannot have both multipliers positive. At $\varepsilon=0$, both may be positive; 预测时使用两者之差。Tube-boundary location does not require a positive coefficient.

背面第 4 栏 | LS-SVM 推导

L01 Least-Squares Support Vector Machine

答卷句： LS-SVM uses squared loss, norm regularization and equality constraints. Training solves a linear system. 以下按**回归**题型填写。

题设： n samples (x_i,y_i), $x_i\in\mathbb{R}^d,\ y_i\in\mathbb{R}$; feature vector $\phi_i\in\mathbb{R}^h$; kernel $K_{ij}=\phi_i^T\phi_j$; $\gamma>0$. **未知量：** $w\in\mathbb{R}^h,\ b\in\mathbb{R}$, residuals $e\in\mathbb{R}^n$, multipliers $\alpha\in\mathbb{R}^n$.

1. 写模型、目标与约束

$$f(x)=w^T\phi(x)+b,\qquad e_i=y_i-f(x_i),$$

$$\boxed{\min_{w,b,e}\ \tfrac12\|w\|^2+\tfrac{\gamma}{2}\sum_{i=1}^n e_i^2}$$

$$\boxed{y_i=w^T\phi_i+b+e_i,\quad i=1,\dots,n.}$$

答卷句： The norm term controls complexity. Squared loss penalizes residuals. A larger γ emphasizes training fit. Equality constraints allow nonzero errors; they do not force interpolation.

2. 构造拉格朗日函数

$$\begin{aligned}\mathcal{L}=&\tfrac12w^Tw+\tfrac{\gamma}{2}\sum_ie_i^2\\&-\sum_i\alpha_i(w^T\phi_i+b+e_i-y_i).\end{aligned}$$

3. 分别求导并置零

$$\partial_w\mathcal{L}=w-\sum_i\alpha_i\phi_i=0,$$

$$\partial_b\mathcal{L}=-\sum_i\alpha_i=0,$$

$$\partial_{e_i}\mathcal{L}=\gamma e_i-\alpha_i=0,$$

$$\partial_{\alpha_i}\mathcal{L}=-(w^T\phi_i+b+e_i-y_i)=0.$$

4. 整理

$$\boxed{w=\sum_i\alpha_i\phi_i,\quad \sum_i\alpha_i=0,\quad e_i=\alpha_i/\gamma}$$

注意： Equality multipliers satisfy $\alpha_i\in\mathbb{R}$; they have no box constraints. 不要把 y_i 加入 $w=\sum_i\alpha_i\phi_i$.

L02 Linear system and prediction

5. 代入 w 与 e_i

$$y_i=\sum_j\alpha_jK_{ij}+b+\gamma^{-1}\alpha_i.$$

6. 按 $[\alpha;b]$ 顺序合并方程

$$\boxed{\begin{bmatrix}K+\gamma^{-1}I&\mathbf{1}\\\mathbf{1}^T&0\end{bmatrix}\begin{bmatrix}\alpha\\b\end{bmatrix}=\begin{bmatrix}y\\0\end{bmatrix}}.$$

$K,I\in\mathbb{R}^{n\times n}$; $\mathbf{1},y,\alpha\in\mathbb{R}^n$; the full system is $(n+1)\times(n+1)$. **两样本照填模板：**

$$\begin{bmatrix}K_{11}+1/\gamma&K_{12}&1\\K_{21}&K_{22}+1/\gamma&1\\1&1&0\end{bmatrix}\begin{bmatrix}\alpha_1\\\alpha_2\\b\end{bmatrix}=\begin{bmatrix}y_1\\y_2\\0\end{bmatrix}.$$

照做： 算所有 $K_{ij}\to$ 对角加 $1/\gamma\to$ 右补一列 1、下补一行 $[1,\dots,1,0]\to$ 右端填 $[y;0]\to$ 消元回代求 α,b。

7. 计算预测

$$\boxed{f(x)=\sum_i\alpha_iK(x_i,x)+b.}$$

8. 回代检查： $\sum_i\alpha_i=0$，且

$$e=y-K\alpha-b\mathbf{1}=\alpha/\gamma.$$

For a linear kernel, $w=X^T\alpha$, where row i of X is x_i^T. **注意：** 对角加 $+1/\gamma$；末行用 $\mathbf{1}^T$。The bias is not regularized; its diagonal entry is 0.

背面第 5 栏 | LS-SVM / 比较

L03 LS-SVM: 完整计算

题设： $x=(0,1),\ y=(0,2),\ K(x,z)=xz,\ \gamma=1$. 求 α,b 与 $f(2)$。

$$K=\begin{bmatrix}0&0\\0&1\end{bmatrix},\qquad \left[\begin{array}{ccc|c}1&0&1&0\\0&2&1&2\\1&1&0&0\end{array}\right].$$

行操作：$R_3\leftarrow R_3-R_1$；交换 R_2,R_3；$R_3\leftarrow R_3-2R_2$。

$$\left[\begin{array}{ccc|c}1&0&1&0\\0&1&-1&0\\0&0&3&2\end{array}\right].$$

回代：$3b=2\Rightarrow b=2/3$; $\alpha_2-b=0\Rightarrow\alpha_2=2/3$; $\alpha_1+b=0\Rightarrow\alpha_1=-2/3$。$w=0\alpha_1+1\alpha_2=2/3$. 由此

$$\boxed{f(x)=\tfrac23x+\tfrac23,\quad f(2)=2.}$$

检查：$f(X)=(2/3,4/3)$;
$e=y-f(X)=(-2/3,2/3)=\alpha/\gamma$; $\sum\alpha=0$。

$$J=\tfrac12(2/3)^2+\tfrac12[(-2/3)^2+(2/3)^2]=2/3.$$

换参数： 若 $\gamma=2$，对角改加 $1/2$，重新解得 $\alpha=(-1,1)$, $b=1/2$, $f(x)=x+1/2$.

答卷句： If $K\succeq0$ and $\gamma>0$, $K+\gamma^{-1}I\succ0$ and the bias system is invertible ($n\ge1$), even for singular K. LS-SVM is usually dense: $\alpha_i=\gamma e_i$. Squared loss is sensitive to outliers.

D06 SVM V.S. SVR V.S. LS-SVM

共同点： All use norm regularization and can use kernels. All require model selection and can overfit.

SVM classification: It predicts classes using a margin criterion. Soft SVM uses linear slack penalties and inequalities. Its dual is a quadratic program (QP), with $0\le\alpha_i\le C$.

$$\hat{y}=\operatorname{sign}\left(\sum_i\alpha_iy_iK(x_i,x)+b\right).$$

SVR: It predicts continuous values. Errors inside the ε-tube are unpenalized; outside errors incur linear loss. Its dual is a QP with $0\le\alpha_i,\alpha_i^*\le C$.

$$f(x)=\sum_i(\alpha_i-\alpha_i^*)K(x_i,x)+b.$$

LS-SVM regression: It uses squared errors, equality constraints and unrestricted multipliers. Training solves a linear system; $f(x)=\sum_i\alpha_iK(x_i,x)+b$.

限制： SVM/SVR can be sparse. Hard SVM needs separability; SVR needs a suitable ε. LS-SVM also supports classification. **判断最大 / 最小：** Hard SVM maximizes geometric margin, equivalently minimizing $\|w\|^2/2$. SVM/SVR dual objectives are maximized. LS-SVM minimizes a convex quadratic objective; its stationarity equations form a linear system.

参数影响： Larger C penalizes margin/tube violations more. Larger γ emphasizes squared-error fit. Larger ε widens the zero-loss tube. None guarantees better test accuracy. Parameters are selected using validation data.

查找 题干关键词 → 知识块

对比题 (compare/difference): M-P/P → D01; P/ADALINE → D02; MLP/RBF → D03; RBF/FLN/ELM → D04; hard/soft、RBF kernel → D05; SVM/SVR/LS → D06。

计算定位： perceptron → P01; LMS → A01; forward/BP/momentum → B01–B03; features → R01/R02; gradient/equations → M02/M03。

背面定位： margin/KKT → S01/S02; slack/C → S03; kernel → K01; tube → V01/V02; LS-SVM/γ → L01–L03。**注意：** 先看输出、损失与约束来定位。“Least squares” 也可能对应 LMS 或 RBF 训练，不要只凭一个关键词选式。